

J KARTHIK

192472136

CSA0911

JAVA

CO4 AT2 Q4

Q4 Student Record

25 marks

Design a GUI application that allows the user to maintain a simple student record (add/update/delete/display) using appropriate layout manager, controls, and event handling.

CODE:

```
File Edit Selection View Go Run Terminal Help < -> Java
J MenuDriven.java 9 • J import javax.swing.*; Untitled-1 • J import javax.swing.*; Untitled-1 •

1 import javax.swing.*;
2 import javax.swing.table.DefaultTableModel;
3 import java.awt.*;
4 import java.awt.event.*;
5 public class StudentRecord extends JFrame implements ActionListener {
6     JLabel l1, l2, l3;
7     JTextField t1, t2, t3;
8     JButton add, update, delete, display, clear;
9     JTable table;
10    DefaultTableModel model;
11    StudentRecord() {
12        setTitle("Student Record Management System");
13        setSize(700, 500);
14        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
15        setLayout(new BorderLayout(10, 10));
16        JLabel title = new JLabel(
17            "STUDENT RECORD MANAGEMENT SYSTEM",
18            JLabel.CENTER
19        );
20        title.setFont(
21            new Font("Arial", Font.BOLD, 20)
22        );
23        add(title, BorderLayout.NORTH);
24        JPanel inputPanel = new JPanel(
25            new GridLayout(3, 2, 10, 10)
26        );
27        l1 = new JLabel("Student ID:");
28        l2 = new JLabel("Student Name:");
29        l3 = new JLabel("Department:");
30        t1 = new JTextField();
31        t2 = new JTextField();
32        t3 = new JTextField();
33        inputPanel.add(l1);
34        inputPanel.add(t1);
35        inputPanel.add(l2);
36        inputPanel.add(t2);
37        inputPanel.add(l3);
38        inputPanel.add(t3);
39        inputPanel.add(t3);
40        add(inputPanel, BorderLayout.WEST);
41        String[] columns = {
42            "Student ID",
43            "Student Name",
44            "Department"
45        };
46        model = new DefaultTableModel(
47            columns, 0
48        );
49        table = new JTable(model);
50        JScrollPane scrollPane =
51            new JScrollPane(table);
52        add(
53            scrollPane,
54            BorderLayout.CENTER
55        );
56        JPanel buttonPanel = new JPanel(
57            new FlowLayout()
58        );
59        add = new JButton("Add");
60        update = new JButton("Update");
61        delete = new JButton("Delete");
62        display = new JButton("Display");
63        clear = new JButton("Clear");
64        buttonPanel.add(add);
65        buttonPanel.add(update);
66        buttonPanel.add(delete);
67        buttonPanel.add(display);
68        buttonPanel.add(clear);
69        add(
70            buttonPanel,
71            BorderLayout.SOUTH
72        );
73        add.addActionListener(this);
74        update.addActionListener(this);
75        delete.addActionListener(this);
76        display.addActionListener(this);
77    }
78}
```

```
File Edit Selection View Go Run Terminal Help  ← → Java

J MenuDriven.java 9  J StudentRecord.java 7  ...  J StudentRecord.java 7 X

J StudentRecord.java > Language Support for Java(TM) by Red Hat > StudentRecord > StudentRecord()
5 public class StudentRecord extends JFrame implements ActionListener {
11     StudentRecord() {
38         inputPanel.add(t3);
39         add(inputPanel, BorderLayout.WEST);
40         String[] columns = {
41             "Student ID",
42             "Student Name",
43             "Department"
44         };
45         model = new DefaultTableModel(
46             columns, rowCount: 0
47         );
48         table = new JTable(model);
49         JScrollPane scrollPane =
50             new JScrollPane(table);
51         add(
52             scrollPane,
53             BorderLayout.CENTER
54         );
55         JPanel buttonPanel = new JPanel(
56             new FlowLayout()
57         );
58         add = new JButton(text: "Add");
59         update = new JButton(text: "Update");
60         delete = new JButton(text: "Delete");
61         display = new JButton(text: "Display");
62         clear = new JButton(text: "Clear");
63         buttonPanel.add(add);
64         buttonPanel.add(update);
65         buttonPanel.add(delete);
66         buttonPanel.add(display);
67         buttonPanel.add(clear);
68         add(
69             buttonPanel,
70             BorderLayout.SOUTH
71         );
72         add.addActionListener(this);
73     }
74 }

J StudentRecord.java > Language Support for Java(TM) by Red Hat > StudentRecord > StudentRecord()
5 public class StudentRecord extends JFrame implements ActionListener {
11     StudentRecord() {
72         add.addActionListener(this);
73         update.addActionListener(this);
74         delete.addActionListener(this);
75         display.addActionListener(this);
76         clear.addActionListener(this);
77         table.getSelectionModel().
78             .addListSelectionListener(e -> {
79                 int row =
80                     table.getSelectedRow();
81                 if (row >= 0) {
82                     t1.setText(
83                         model.getValueAt(
84                             row, column: 0
85                         ).toString()
86                     );
87                     t2.setText(
88                         model.getValueAt(
89                             row, column: 1
90                         ).toString()
91                     );
92                     t3.setText(
93                         model.getValueAt(
94                             row, column: 2
95                         ).toString()
96                     );
97                 }
98             });
99         setLocationRelativeTo(c: null);
100         setVisible(b: true);
101     }
102 }
103 public void actionPerformed(ActionEvent e) {
104     if (e.getSource() == add) {
105         if (t1.getText().isEmpty())
106             ...
107     }
108 }

Ln 16, Col 35  Spaces: 8  UTF-8  LF  { } Java  Go Live  Go Live  Prettier
```

```
File Edit Selection View Go Run Terminal Help  ← → Java

J MenuDriven.java 9  J StudentRecord.java 7  ...  J StudentRecord.java 7 X

J StudentRecord.java > Language Support for Java(TM) by Red Hat > StudentRecord > StudentRecord()
5 public class StudentRecord extends JFrame implements ActionListener {
103 public void actionPerformed(ActionEvent e) {
106     if (t1.getText().isEmpty()
107         || t2.getText().isEmpty()
108         || t3.getText().isEmpty()) {
109
110         JOptionPane.showMessageDialog(
111             this,
112             message: "Please enter all details."
113         );
114
115         return;
116     }
117
118     model.addRow(
119         new Object[]{
120             t1.getText(),
121             t2.getText(),
122             t3.getText()
123         }
124     );
125
126     JOptionPane.showMessageDialog(
127         this,
128         message: "Student added successfully!"
129     );
130
131     clearFields();
132 }
133 else if (e.getSource() == update) {
134
135     int row =
136         table.getSelectedRow();
137
138     if (row == -1) {
139
140         JOptionPane.showMessageDialog(
141             this,
142             message: "Please select a student."
143         );
144
145         return;
146     }
147
148     model.setValueAt(
149         t1.getText(), row, column: 0
150     );
151
152     model.setValueAt(
153         t2.getText(), row, column: 1
154     );
155
156     model.setValueAt(
157         t3.getText(), row, column: 2
158     );
159
160     JOptionPane.showMessageDialog(
161         this,
162         message: "Student updated successfully!"
163     );
164
165     clearFields();
166 }
167 else if (e.getSource() == delete) {
168
169     int row =
170         table.getSelectedRow();
171
172     if (row == -1) {
173
174         JOptionPane.showMessageDialog(
175             this,
```

```
File Edit Selection View Go Run Terminal Help
J MenuDriven.java 9 J StudentRecord.java 7 J StudentRecord.java 7 X

J StudentRecord.java > Language Support for Java(TM) by Red Hat > StudentRecord > StudentRecord()
5 public class StudentRecord extends JFrame implements ActionListener {
103 public void actionPerformed(ActionEvent e) {
176     message: "Please select a student."
177 }
178
179 return;
180 }
181
182 model.removeRow(row);
183
184 JOptionPane.showMessageDialog(
185     this,
186     message: "Student deleted successfully!"
187 );
188
189 clearFields();
190 }
191 else if (e.getSource() == display) {
192
193     if (model.getRowCount() == 0) {
194
195         JOptionPane.showMessageDialog(
196             this,
197             message: "No student records available."
198         );
199
200     } else {
201
202         JOptionPane.showMessageDialog(
203             this,
204             message: "Student records are displayed in the table."
205         );
206     }
207 }
208 else if (e.getSource() == clear) {
209
210     clearFields();
211 }
212 }
213
214 void clearFields() {
215
216     t1.setText(t: "");
217     t2.setText(t: "");
218     t3.setText(t: "");
219
220     table.clearSelection();
221 }
222
223 Run main | Debug main | Run | Debug
224 public static void main(String[] args) {
225     new StudentRecord();
226 }
```

OUTPUT:

Student ID:

Student Name:

Department:

Student ID	Student Name	Department
192472136	Karthik	CSE - AI
192425143	Madhavan	CSA - AI
192424294	Pavan	CSE - AIDS

Add

Update

Delete

Display

Clear